

Anemia and Red Blood Cell Transfusions

A conscientious man would be cautious how he dealt in blood.

Edmund Burke ([a](#))

As many as 50% of ICU patients are transfused with red blood cells ([1](#)), and in a large majority (90%) of cases, these transfusions are used to boost the hemoglobin level in nonbleeding patients ([2](#)). The practice of transfusing red blood cells (RBCs) based on the hemoglobin level creates a question about the correlation between the hemoglobin level in blood and the adequacy of tissue oxygenation (since the goal of RBC transfusions should be to promote tissue oxygenation). This is a relevant issue because there are considerable risks associated with RBC transfusions, and there is a concern that these risks may outweigh the risks of the anemia they are meant to correct ([3](#)).

This chapter focuses on the influence of anemia and RBC transfusions on tissue oxygenation. It begins by describing the causes and consequences of anemia in the critically ill, and then presents the indications, physiological effects, and adverse consequences associated with RBC transfusions. Current practices that lack a rational or scientific basis are highlighted, to address the need for a more judicious use of RBC transfusions.

ANEMIA IN THE ICU

Anemia is almost universal in the ICU; i.e., it is present on admission in two-thirds of the patients, and the prevalence increases to 97% after one week in the ICU ([4](#)). However, the standard definition of anemia can be problematic in ICU patients, as explained next.

Definition of Anemia

Anemia is defined as a *decrease in the oxygen carrying capacity of blood*. The most accurate measure of the O₂ carrying capacity of blood is the *red cell mass*, which is the volume of circulating red blood cells, and is measured using chromium-tagged erythrocytes. This measurement is not typically used in clinical settings; instead, the hematocrit (Hct) and

hemoglobin concentration are used as clinical measures of O₂ carrying capacity. (Reference ranges for these red cell parameters are shown in [Table 12.1](#)) *The clinical definition of anemia is a hemoglobin (Hb) concentration <13 g/dL (Hct <40%) in males, and <12 g/dL (Hct <38%) in females (4).* Since this definition is based on a concentration, it will be influenced by the plasma volume.

TABLE 12.1 **Reference Ranges for Red Cell Parameters in Adults**

Hemoglobin (Hb) Males: 13.5–18 g/dL Females: 12–16 g/dL*	Mean Cell Volume Males: 80–100 x 10 ⁻¹⁵ /L Females: same
Hematocrit (Hct) Males: 40–54% Females: 38–47%	Red Blood Cell Count Males: 4.6–6.2 x 10 ¹² /L Females: 4.2–5.4 x 10 ¹² /L
Red Cell Mass Males: 26 mL/kg Females: 24 mL/kg	Reticulocyte Count Males: 25–75 x 10 ⁹ /L Females: same

*Normal range is 1 g/dL lower after the first trimester of pregnancy.

Sources: (1) Walker RH (ed). Technical Manual of the American Association of Blood Banks. 10th ed. Arlington, VA: American Association of Blood Banks, 1990:649–650; (2) Hillman RS, Finch CA. Red cell manual. 6th ed. Philadelphia: FA Davis, 1994;46.

Influence of Plasma Volume

The confounding influence of the plasma volume on a clinical measure of O₂ carrying capacity (Hct) is demonstrated in [Figure 12.1](#). The panel on the left shows the postural changes in Hct in a group of healthy adults (5). Note that there is a 4.1% absolute decrease in Hct in the supine position. This is the result of a decrease in the hydrostatic pressure within the capillaries in the legs (due to loss of the gravitational effect), which promotes the movement of interstitial fluid into the bloodstream and increases the plasma volume (by 420 mL in this study). The panel on the right shows the effect of a bolus infusion of isotonic saline (20 mL/kg), which causes a similar (4.2%) decrease in Hct (6). In both cases, the decrease in Hct is equivalent to the change produced by the loss of one unit of red blood cells.

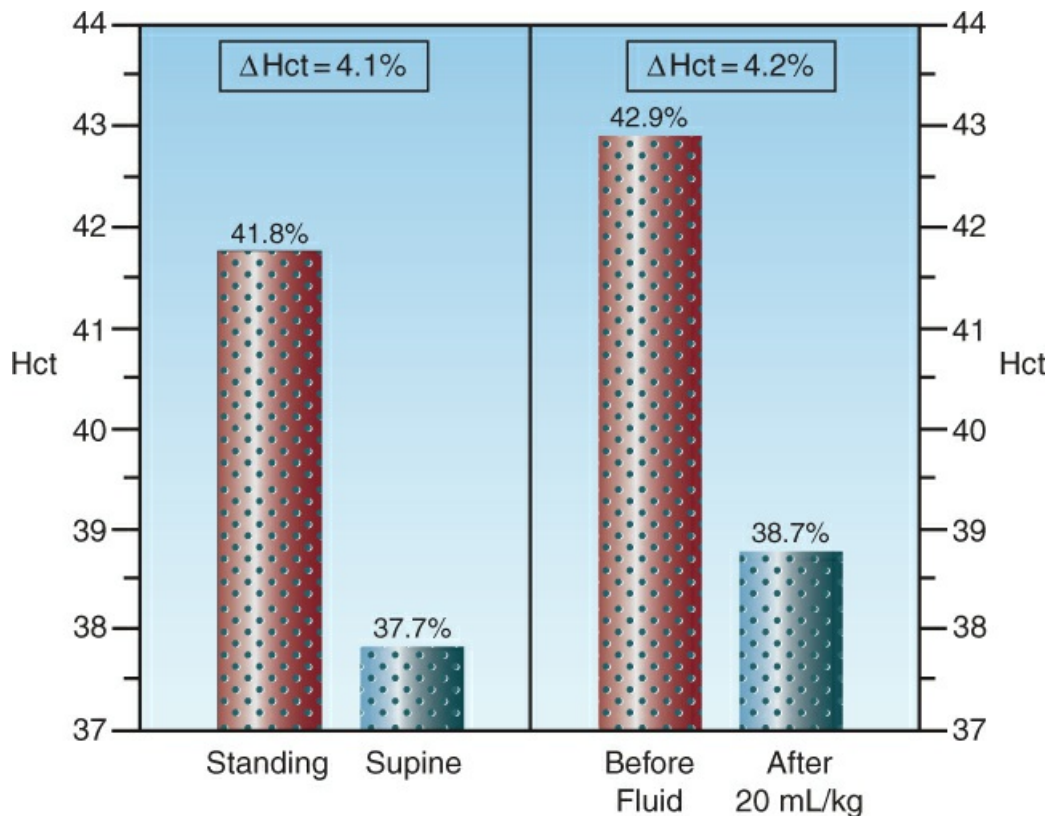


FIGURE 12.1 The influence of plasma volume on the hematocrit (Hct). The panel on the left shows the postural decrease in Hct in the supine position, which is explained in the text. The panel on the right shows the decrease in Hct 90 minutes after a bolus infusion of isotonic saline at 20 mL/kg. In both cases, the absolute decrease in Hct is equivalent to the change produced by loss of one unit of blood (500 mL). From References 6 and 7.

An increased plasma volume is common in critically ill patients, as emphasized in [Chapter 11](#) (see [Figure 11.3](#)), and this means that *the Hct and Hb concentration will overestimate the prevalence and severity of anemia in ICU patients*. Clinical studies have confirmed that the Hct and Hb concentration are unreliable measures of anemia in critically ill patients (7,8). Unfortunately, these are the measures used in all the clinical studies evaluating anemia and erythrocyte transfusions in critically ill patients.

ICU-Related Anemia

Two conditions can add to the prevalence and severity of anemia during the ICU stay: systemic inflammation, and repeated phlebotomy for laboratory studies.

Inflammation

Systemic inflammation is accompanied by iron sequestration in tissue macrophages, which is attributed to the actions of cytokines, and also to the hepatic release of *hepcidin* (9), a small peptide that reduces plasma iron levels by promoting iron sequestration in tissue macrophages (9). The sequestered iron cannot be transferred to developing red blood cells, and this results in a hypochromic microcytic anemia that resembles iron-deficiency anemia. (This is the same mechanism for the “anemia of chronic disease”). The plasma iron profile in the *anemia of*

inflammation includes a decrease in plasma iron, total iron binding capacity, and transferrin levels, which can be indistinguishable from iron deficiency, but there is an increase in plasma ferritin levels.

PLASMA FERRITIN: The ferritin level in plasma is used to evaluate tissue iron stores, and thus can distinguish between iron deficiency anemia (plasma ferritin <30 µg/L) and the anemia of inflammation (plasma ferritin >100 µg/L) (9).

Phlebotomy

An average of 40–70 mL of blood is withdrawn daily in ICU patients to perform laboratory tests (4,10), and the cumulative blood loss can reach 500 mL (one unit of whole blood) after one week.

The daily phlebotomy volume can be reduced by reinfusing the initial sample on a blood draw; i.e., when blood is withdrawn through vascular catheters for laboratory testing, the initial aspirate is discarded to eliminate interference from intravenous fluid in the lumen of the catheter. The volume of discarded blood is typically about 5 mL for each laboratory blood draw, and returning this blood to the patient can reduce the daily phlebotomy volume by 50% (11).

The Physiology of Anemia

Anemia elicits two responses that help to preserve tissue oxygenation: (1) an increase in cardiac output (the result of a decrease in blood viscosity), and (2) an increase in O₂ extraction from capillary blood.

Blood Viscosity

Viscosity is defined as the resistance of a fluid to changes in flow rate (12), and has also been described as the “gooiness” of a fluid (13). The viscosity of blood is the result of cross-linking of erythrocytes by plasma fibrinogen, and the principal determinant of whole blood viscosity is the concentration of erythrocytes (the hematocrit). The influence of hematocrit on blood viscosity is shown in Table 12.2. Note that blood viscosity can be expressed in absolute or relative terms (relative to water). The viscosity of plasma (zero hematocrit) is only slightly higher than that of water, while the viscosity of whole blood at a normal hematocrit (45%) is about 3 times greater than plasma and about 4 times greater than water. Thus, anemic blood flows much more readily than normal blood.

TABLE 12.2 Relationship Between Hematocrit and Blood Viscosity		
Hematocrit (%)	Relative Viscosity (water = 1)	Absolute Viscosity (centipoise)
0	1.4	—
10	1.8	1.2
20	2.1	1.5
30	2.8	1.8
40	3.7	2.3

50	4.8	2.9
60	5.8	3.8

From Documenta Geigy Scientific Tables. 7th ed. Basel: Documenta Geigy, 1966:557–8.

RESISTANCE TO FLOW: The influence of blood viscosity on peripheral blood flow can be described with the Hagen-Poiseuille equation shown below (14), which is also used in [Chapter 1](#) to describe flow through vascular catheters.

$$Q = \Delta P \times (\pi r^4 / 8\mu L) \quad (12.1)$$

This equation states that flow (Q) through a blood vessel is directly related to the pressure gradient along the vessel (ΔP) and the fourth power of the radius (r) of the vessel, and is inversely related to the length (L) of the vessel and the viscosity (μ) of the fluid. The final term in the equation is the reciprocal of resistance (1/R), so resistance to flow can be described as:

$$R = 8\mu L / \pi r^4 \quad (12.2)$$

This equation indicates that flow resistance in the peripheral circulation is directly related to blood viscosity.

Cardiac Output

The effects of anemia on blood viscosity are responsible for the increase in cardiac output shown in [Figure 12.2 \(15\)](#). Note that the increase in cardiac output is proportionally greater than the decrease in hematocrit. This is because blood is a *non-Newtonian fluid*, which means that viscosity is influenced by blood flow; i.e., an increase in blood flow will reduce viscosity. (In a teleological sense, this property would be advantageous in helping to reduce blood loss through a punctured blood vessel; i.e., blood loss through a puncture would lead to a decrease in local blood flow, which would then “thicken” the blood and reduce the tendency for further blood loss.) The increase in cardiac output helps to maintain systemic O₂ transport in the face of a declining hematocrit.

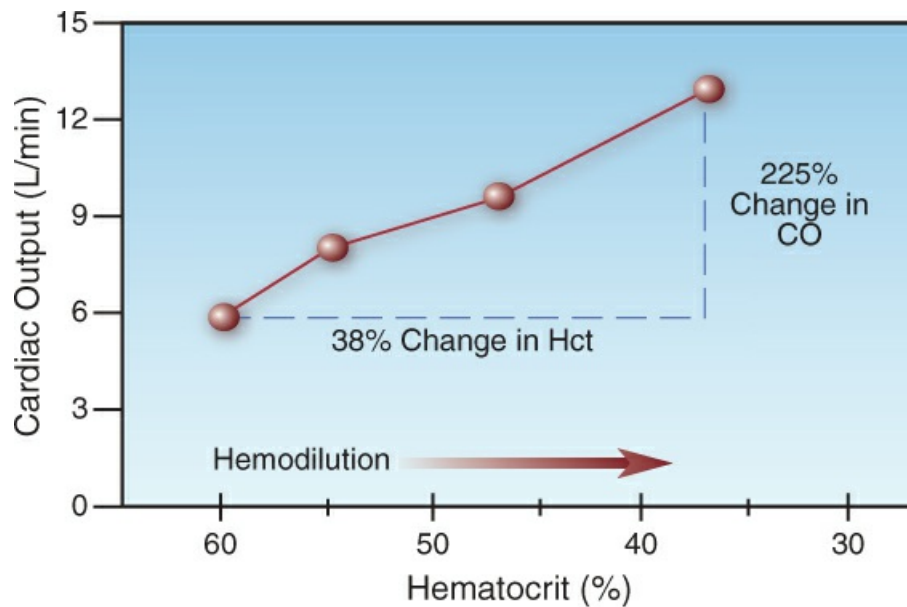


FIGURE 12.2 The influence of progressive isovolemic hemodilution on cardiac output in a patient with polycythemia. From Reference 15.

Systemic Oxygenation

The influence of progressive anemia on measures of systemic oxygenation are shown in [Figure 12.3 \(16\)](#). The principal findings are explained using the following relationships between O_2 uptake (VO_2), O_2 delivery (DO_2), and O_2 extraction:

$$VO_2 = DO_2 \times O_2 \text{ Extraction} \quad (12.3)$$

The upper panel in [Figure 12.3](#) shows that, despite the increase in cardiac output in anemia, the decrease in hemoglobin results in an overall decrease in systemic O_2 delivery. However, there is a compensatory increase in O_2 extraction, and this maintains a constant O_2 uptake (shown in the lower panel). However, when the hematocrit falls below 10%, the increase in O_2 extraction is no longer able to fully compensate for the decrease in O_2 delivery, and the O_2 uptake then begins to fall. When this occurs, the lactate level begins to rise, indicating inadequate tissue oxygenation. Note that the maximum O_2 extraction is about 50 %, and this marks the threshold for impaired tissue oxygenation.

There are two features in [Figure 12.3](#) that deserve emphasis:

- . Anemia did not impair tissue oxygenation until the hematocrit fell to 10% (equivalent to a Hb concentration of 3 g/dL), indicating that even the most severe anemias are tolerated (thanks to the compensatory increase in O_2 extraction).
- . An O_2 extraction that has increased to 50% marks the point where tissue oxygenation is threatened, and this point could serve as an indication for RBC transfusions. (More on this a little later).

TOLERANCE TO SEVERE ANEMIA: Animal studies have shown that, when the intravascular volume

is maintained, hematocrits as low as 5 to 10% (Hb = 1.5 to 3 g/dL) do not adversely affect tissue oxygenation (16–18), even in awake animals breathing room air (18). Human studies of the lowest tolerable hematocrit or hemoglobin mostly involve Jehovah’s Witness (JW) patients. In a large study that included 300 postoperative JW patients, 50% of the patients survived with a Hb level of 2–3 g/dL, and 75% of the patients survived with a Hb level of 3–4 g/dL (19). Another study of progressive hemodilution in healthy adults showed that Hb levels of 5 g/dL produced no apparent harm (20). Thus, the available studies suggest that Hb levels down to 5.0 are safe, and individual patients can tolerate Hb levels as low as 2–3 g/dL!

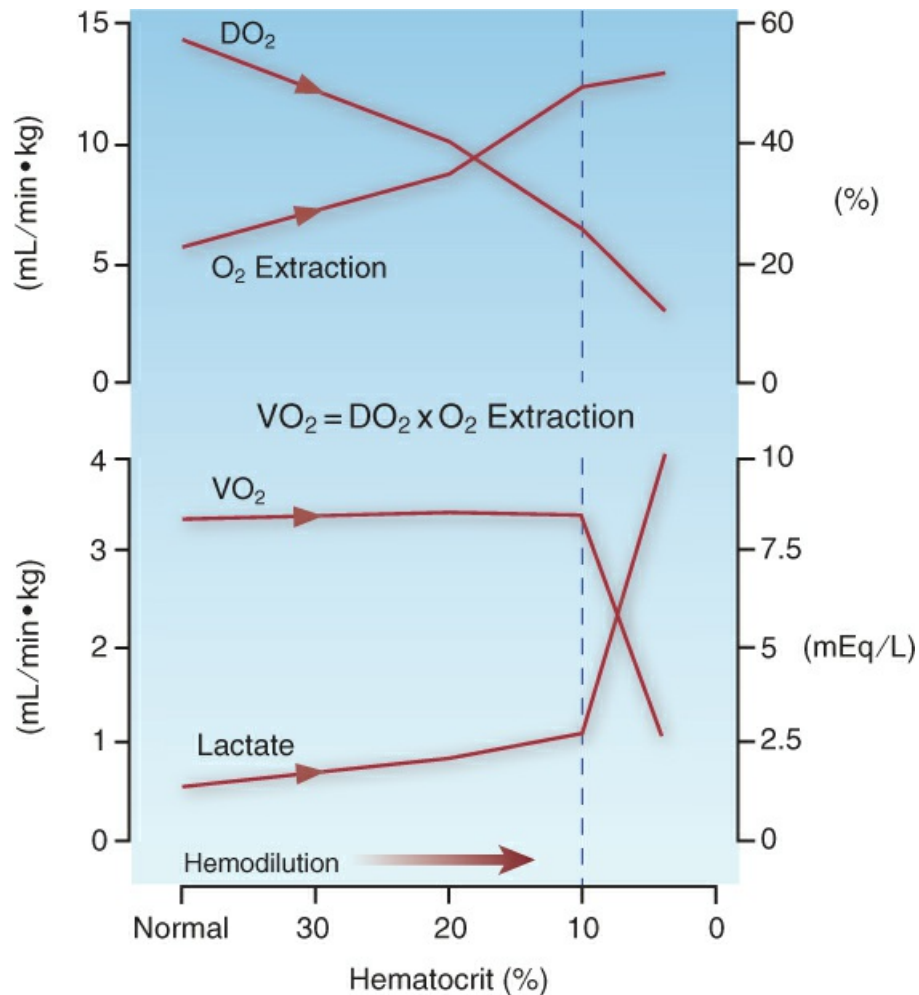


FIGURE 12.3 The influence of progressive isovolemic anemia on measures of systemic oxygenation in nonhuman primates. DO₂ = O₂ delivery, VO₂ = O₂ uptake. See text for explanation. Data from Reference 16.

THE TRANSFUSION TRIGGER

In 1942, a hemoglobin (Hb) level of ≤10 g/dL was recommended as an indication for RBC transfusions (21), and this was the standard *transfusion trigger* for the next 60 years, until concern about the risks associated with RBC transfusions prompted studies which showed that outcomes were not adversely affected if Hb levels were allowed to decrease to 7 g/dL (22,23). A

recent review of 48 clinical studies confirmed that outcomes were similar (including cardiac events) when Hb levels were maintained at 7–8 g/dL instead of 9–10 g/dL, and that the lower Hb threshold for transfusion resulted in a 41% reduction in the consumption of RBC products (24).

Guidelines

There are 21 clinical practice guidelines for RBC transfusions in critically ill patients who are not actively bleeding (for the recommendations of each, see Ref. 25), and the majority of the recommendations state the following:

- . The threshold for RBC transfusions is a Hb <7 g/dL for critically ill patients who are hemodynamically stable, and do not have coronary artery disease. This includes post-cardiac surgery patients and patients with septic shock.
- . The transfusion threshold should be a Hb <8 g/dL for patients with coronary artery disease, and for patients undergoing cardiac surgery or orthopedic surgery.

Compliance with these guidelines has been poor; e.g., one survey of close to 5,000 RBC transfusions revealed that 60% of transfusions did not follow the recommendations of the guidelines (26).

What's Wrong?

There are two problems with the use of the Hb concentration as a transfusion trigger:

- . The Hb concentration provides no information about the adequacy of tissue oxygenation, and since the goal of RBC transfusions is to promote tissue oxygenation, transfusions based on the Hb concentration are given without evidence of need or benefit.
- . Acute decreases in the Hb concentration can be a dilutional effect, and this can result in inappropriate RBC transfusions.

Some clinical practice guidelines have recommended abandoning the Hb level as a transfusion trigger and adopting more physiologic measures of tissue oxygenation (27), like the ones described next (27).

Oxygen Extraction

As described earlier (and shown in Figure 12.3) anemia elicits a compensatory increase in O_2 extraction, which helps to maintain a constant rate of O_2 uptake into tissues. However, the O_2 extraction reaches a maximum of about 50%, and beyond this point, further decreases in Hb will result in a decrease in O_2 uptake, which is a marker of impaired tissue oxygenation. Therefore, *an O_2 extraction of 50% can be used as a transfusion trigger* because it identifies the threshold for impaired tissue oxygenation (28). The O_2 extraction is roughly equivalent to the difference between the arterial and central venous O_2 saturation, ($SaO_2 - ScvO_2$), and this can be monitored continuously using pulse oximetry (for the SaO_2) and a central venous “oximetry” catheter (for the $ScvO_2$).

When the SaO_2 is close to 100%, the $ScvO_2$ can be used alone as a transfusion trigger; e.g.,

an $\text{ScvO}_2 < 70\%$ has been proposed as a transfusion trigger (29).

Unfortunately, despite the physiological advantages of these approaches, they have received little attention.

RED BLOOD CELL TRANSFUSIONS

Whole blood is stored only on request, and is otherwise separated into its component parts; i.e., red blood cells (RBCs), platelets, plasma, and cryoprecipitate. This practice allows each unit of donated blood to serve multiple transfusion needs.

Preparations

The RBC preparations that are available for transfusion are shown in Table 12.3.

TABLE 12.3 Red Blood Cell Preparations	
Preparation	Features
Packed RBCs	1. Each unit has a volume of 350 mL and hematocrit of about 60%. 2. Contains leukocytes and residual plasma (15–30 mL per unit) 3. Can be stored for 42 days with appropriate additives
Leukocyte-reduced RBCs	1. Donor RBCs are passed through specialized filters to remove most of the leukocytes. This reduces the risk of nonhemolytic febrile reactions. 2. Indicated for patients with a history of (nonhemolytic) febrile transfusion reactions
Washed RBCs	1. Packed RBCs are saline washed to remove residual plasma, which reduces the risk of hypersensitivity reactions. 2. Indicated for patients with a history of transfusion-related allergic reactions, and in patients with IgA deficiency, who are at risk for transfusion-related anaphylaxis

From Reference 30.

Packed RBCs

The erythrocyte fraction of donated blood is placed in a preservative fluid and stored at 1 to 6° C. Newer preservative solutions contain adenine, which helps to maintain ATP levels in stored erythrocytes, and allows storage of donor erythrocytes for up to 42 days (30). Each unit of concentrated erythrocytes, known as *packed red blood cells* (packed RBCs), has a hematocrit of about 60% and a volume of about 350 mL, which includes about 30 – 50 mL of residual plasma, and a considerable number of leukocytes (30).

Leukocyte-Reduced RBCs

The leukocytes in packed RBCs can trigger an antibody response in the recipient after repeated transfusions, and this is responsible for febrile nonhemolytic transfusion reactions (see later). To reduce the risk of this reaction, donor RBCs are passed through specialized filters to remove most of the leukocytes. This is performed routinely in many blood banks, but universal leukocyte reduction has yet to be adopted in the United States. Leukocyte-reduced RBCs are recommended for patients with prior febrile nonhemolytic transfusion reactions (30).

Washed RBCs

Donor RBCs can be washed with isotonic saline to remove residual plasma. This reduces the risk of hypersensitivity reactions caused by prior sensitization to plasma proteins in donor blood. Washed RBC preparations are recommended for patients with a history of hypersensitivity reactions to blood transfusions, and for patients with immunoglobulin A deficiency, who have an increased risk of transfusion-related anaphylaxis (30). Saline washing does not effectively remove leukocytes.

Infusing Packed RBCs

Packed RBCs (PRBCs) have a high viscosity (due to the Hct of 60%) and thus have a sluggish flow rate. As a result, saline is often added to PRBC transfusions to increase the flow rate and reduce infusion time. The data in Table 12.4 shows the infusion time for one unit of PRBCs through 18- and 20-gauge peripheral catheters, and the influence of dilution with 100 mL isotonic saline (31). Note that the infusion time is almost 2 hrs when the undiluted RBCs are infused through a 20-gauge catheter, and the infusion time is reduced to almost one hour with an 18-gauge catheter. Also note that the RBCs infuse six times faster when diluted with saline. The recommended infusion time is typically 2 hrs per unit of packed RBCs for hemodynamically stable patients (30), so the saline dilution is probably not necessary when PRBCs are infused to correct anemia.

TABLE 12.4 Gravity-Driven Infusion of Packed red Blood Cells			
		Infusion Time	
		20-G Catheter*	18-G Catheter
1 Unit Undiluted	350 mL	117 min	70 min
1 Unit + 100 mL Isotonic Saline	450 mL	19 min	12 min

* G = gauge size. Catheter length 1.2 inches for all infusions. From Reference 31.

Blood Filters

Standard blood filters (pore size 170 to 260 microns) are required for the transfusion of all blood products (30). These filters trap blood clots and other debris, but they do not trap leukocytes, and are not effective for leukocyte reduction (30). These filters can become an impediment to flow as they collect trapped debris, and sluggish infusion rates are an indication to replace filters.

Effects on Systemic Oxygenation

In an average-sized adult, one unit of packed RBCs is expected to raise the hemoglobin concentration and hematocrit by 1 g/dL and 3%, respectively (30). The effects of RBC transfusions on measures of systemic oxygenation are shown in Figure 12.4 The data in this figure is from a group of postoperative patients with severe normovolemic anemia (Hb <7 g/dL) who were transfused with 1 – 2 units of packed RBCs to raise the Hb above 7 g/dL. The RBC transfusions increased the mean Hb concentration from 6.4 to 8 g/dL (25% increase), and there was a similar increase in O₂ delivery (DO₂). However, the systemic O₂ uptake (VO₂) was

unchanged. The constant VO_2 in the face of an increased DO_2 indicates that O_2 extraction was reduced by the RBC transfusions, as predicted by Equation 12.3. These changes in DO_2 and O_2 extraction are the reverse of the changes produced by anemia, as shown in Figure 12.3.

No Improvement in Tissue Oxygenation

The lack of an effect on VO_2 in Figure 12.4 indicates that *RBC transfusions do not enhance tissue oxygenation*. This has been confirmed in several clinical studies (32–35), and prolonged storage of RBCs can actually *impair* tissue oxygenation after transfusion (36). These studies have prompted the following statement in a clinical practice guideline for red blood cell transfusions (27):

“RBC transfusion should not be considered an absolute method to improve tissue oxygenation in critically ill patients.”

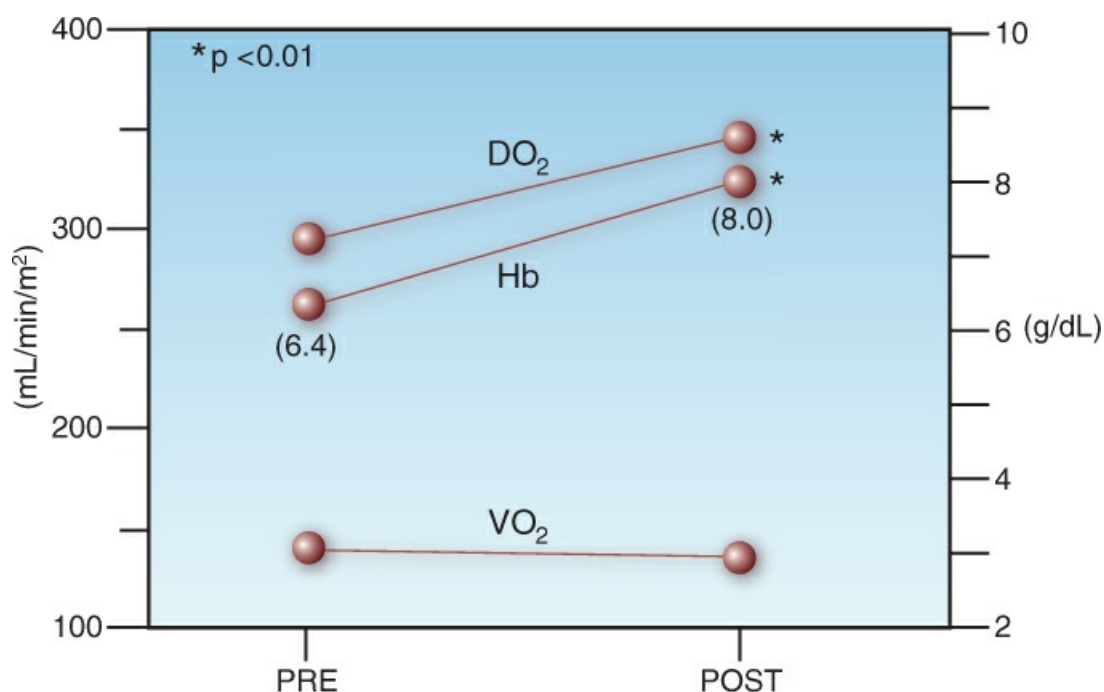


FIGURE 12.4 Effect of RBC transfusions (1–2 units packed RBCs) to correct severe anemia ($\text{Hb} < 7\text{g/dL}$) on systemic oxygen delivery (DO_2), and oxygen uptake (VO_2) in 11 postoperative patients. Data points represent mean values for each parameter. Numbers in parentheses indicate the mean hemoglobin concentrations before and after transfusion. Data from personal observations.

COMMENT: The inability of RBC transfusions to enhance tissue oxygenation raises serious questions about the practice of transfusing RBCs simply to boost the Hb level. This concern is especially relevant in light of the multiple risks associated with RBC transfusions, which are described in the next section.

TRANSFUSION RISKS

The spectrum of adverse events associated with RBC transfusions is shown in Table 12.5, along

with the risk of each event per unit of RBCs transfused (37–42). Note that transfusion errors are much more frequent than the feared transmission of infectious agents. The following is a brief description of the principal transfusion reactions.

TABLE 12.5 Adverse Events Associated with RBC Transfusions (per units transfused)	
Immune-Related	Others
Nonhemolytic fever (1:60) Hypersensitivity Reactions: Urticaria (1:100) Anaphylaxis (1:1,000) Anaphylactic shock (1:50,000) Acute lung injury (1:12,000) Nosocomial Infections (?) Acute hemolysis (1:35,000) Fatal hemolysis (1:1.9 million)	Circulatory Overload (1:100) [†] Transmitted Infections: Bacterial (1:500,000) Hepatitis B virus (1:1.2 million) Hepatitis C virus (1:1.5 million) HIV (1:1.5 million) Transfusion Errors: Wrong person transfused (1:15,000) Incompatible transfusion (1:33,000)

[†]Estimated risk per recipients rather than units. From References 37–42.

Acute Hemolytic Reactions

Acute hemolytic reactions are prompted by the transfusion of RBCs that are ABO-incompatible with the recipient. When this occurs, antibodies in recipient blood bind to ABO antigens on the donor RBCs, and the ensuing lysis of donor RBCs triggers a systemic inflammatory response that can be accompanied by hypotension and multiorgan failure. *These reactions are usually the result of human error (43).*

Clinical Features

The hallmark of acute hemolytic reactions is the abrupt onset of fever, dyspnea, chest pain, low back pain, and hypotension within minutes after starting the transfusion. Severe reactions are accompanied by a consumptive coagulopathy and progressive multiorgan dysfunction.

Management

- . If a hemolytic reaction is suspected, STOP the transfusion immediately and verify that the correct blood was given to the correct patient. It is imperative to stop the transfusion as soon as possible because the severity of hemolytic reactions is a function of the volume of blood transfused (39).
- . If the donor blood is correctly matched to the patient, an acute hemolytic reaction is unlikely. However, the blood bank must be notified, and they will ask for blood samples to perform a plasma-free hemoglobin determination (for evidence of intravascular hemolysis) and a direct Coomb's test (for evidence of the anti-ABO antibody).
- . If an acute hemolytic reaction is confirmed, the management is general supportive care, with volume support and vasopressors for hypotension, and ventilatory support for respiratory distress. These reactions are rarely fatal (see Table 12.5).

Febrile Nonhemolytic Reactions

Febrile reactions that are unrelated to hemolysis are the most common adverse reactions to RBC transfusions (see [Table 12.5](#)). These reactions are characterized by a temperature elevation $>1^{\circ}\text{C}$ (1.8°F) that occurs during, or up to 6 hours after, the transfusion, and there is no other apparent cause (e.g., acute hemolytic reaction) (40). The culprit is the presence of antileukocyte antibodies in recipient blood that react with antigens on leukocytes in donor blood. This triggers the release of endogenous pyrogens from phagocytes, which is the source of the fever. This reaction typically occurs in patients who have received prior transfusions, and in multiparous women. Transfusion of leukocyte-reduced RBCs reduces, but does not eliminate, the risk of this reaction (40).

Clinical Features

The fever does not usually appear in the first hour after the start of the transfusion (unlike the fever associated with acute hemolytic reactions), but it can be accompanied by rigors and chills.

Management

- . The initial approach to transfusion-related fever is the same as described for hemolytic transfusion reactions, even though the fever may not appear until after the transfusion is completed. The diagnosis is confirmed by excluding the presence of hemolysis with the tests described previously.
- . The blood bank will perform a Gram stain on the donor blood, and may request blood cultures from the recipient.
- . More than 75% of nonhemolytic fevers will not recur with subsequent transfusions (44). Therefore, no special precautions are needed for future transfusions. If a second febrile reaction occurs, leukocyte-reduced RBCs are advised for all subsequent transfusions.

Hypersensitivity Reactions

Hypersensitivity reactions are the result of sensitization to plasma proteins in donor blood from prior transfusions. Patients with IgA deficiency are prone to hypersensitivity transfusion reactions, and prior exposure to plasma products is not required. The most common hypersensitivity reaction is urticaria, which is reported in one of every 100 units transfused (41). More severe anaphylactic reactions (e.g., bronchospasm) are much less common, and anaphylactic shock is rare.

Clinical Features

The usual manifestation is mild urticaria that appears during the transfusion and is not accompanied by fever. The abrupt onset of dyspnea and wheezing during a transfusion could be a sign of an anaphylactic reaction, and hypotension from anaphylactic shock can be mistaken for an acute hemolytic reaction.

Management

- . Mild urticaria without fever does not require interruption of the transfusion. However, the

popular practice is to stop the transfusion temporarily and administer an antihistamine for symptom relief (e.g., diphenhydramine, 25–50 mg PO, IM, or IV).

- . Severe anaphylactic reactions should be managed as described in [Chapter 17](#). The transfusion should be stopped immediately if severe anaphylaxis is suspected.
- . For patients who experience an allergic reaction, washed RBCs should be used for all future transfusions, and for patients who experience an anaphylactic reaction, future transfusions should be avoided unless absolutely necessary.
- . Patients who develop hypersensitivity reactions should be tested for an underlying IgA deficiency.

Acute Lung Injury

The condition known as *transfusion-related acute lung injury* (TRALI) is an inflammatory lung injury that resembles the acute respiratory distress syndrome (ARDS), and is associated with RBC and platelet transfusion. It is uncommon (one case per 12,000 RBC transfusions), but is considered the leading cause of transfusion-related deaths, and has a mortality rate that approaches 50% ([45](#)).

Etiology

In 80% of cases, TRALI is the result of antibodies in donor blood that bind to antigens on circulating neutrophils in the recipient ([45](#)). This triggers neutrophil activation, and the activated neutrophils become sequestered in pulmonary capillaries and migrate into the lungs to produce the inflammatory injury. Antibodies are not involved in about 20% of cases, but the responsible agent has not been identified.

Clinical Features

TRALI presents with acute hypoxemic respiratory failure that often appears in the first hour after the transfusion begins, but can appear anytime in the first 6 hours after the start of the transfusion ([46](#)). Fever is common, and the chest x-ray can eventually look like the one in [Figure 12.5](#), with extensive infiltration in both lungs that is indistinguishable from ARDS (which is described in [Chapter 24](#)). The respiratory failure can be severe at the outset, and often requires mechanical ventilation.

TRALI can be difficult to distinguish from the hydrostatic pulmonary edema in transfusion-associated circulatory overload (described next), which can appear in the same time frame after the onset of transfusion and can even be accompanied by fever ([45](#)). Evidence of systemic inflammation (elevated C-reactive protein) will support the diagnosis of TRALI, along with the absence of heart failure or fluid overload ([45](#)).

Management

- . If the transfusion is not completed, it should be stopped at the first signs of respiratory difficulty. The blood bank should be notified for all cases of suspected TRALI. (Assays for antileukocyte antibodies are available, but are not routinely used).
- . The management of TRALI is supportive, and is similar to the management of ARDS described in [Chapter 24](#).

- . There are no firm recommendations regarding future transfusions in patients who develop TRALI. Some recommend using washed RBCs to remove antibodies from donor blood, but the effectiveness of this measure is unknown.

Hydrostatic Pulmonary Edema

The most common pulmonary complication of red blood cell transfusions is a condition known as *transfusion-associated circulatory overload* (TACO), which has a reported incidence of 1% (45,46). The hallmark of this condition is acute hydrostatic pulmonary edema that appears in the first 6 hours after the transfusion begins. The resulting acute hypoxemic respiratory failure often requires mechanical ventilation, and the reported mortality rate is 20% (46).



FIGURE 12.5 Portable chest film from a patient with transfusion-related acute lung injury. Note the homogeneous pattern of infiltration in the lungs, which is a characteristic of inflammatory lung injury.

Etiology

There is no correlation between the number of transfused units and the appearance of TACO (46), but the condition appears more frequently in patients with pre-existing heart failure and renal failure, who are prone to fluid overload (45,46). It is also possible that factors other than the infusion volume are operative in this condition.

Clinical Presentation

The clinical presentation can be indistinguishable from TRALI, with acute onset of hypoxemic respiratory failure in the first 6 hours after the transfusion begins, and a chest x-ray that shows pulmonary edema. Patients are febrile in 30% of cases (45), and mechanical ventilation is required in three-quarters of patients (46). Conditions that favor the diagnosis of TACO include pre-existing heart failure, evidence of fluid overload, or underlying renal insufficiency.

Management

- . The transfusion should be stopped if not completed, and the blood bank should be notified.
- . An intravenous loop diuretic is appropriate. Otherwise, the care is supportive (e.g., with mechanical ventilation, if needed), similar to the management of TRALI.
- . There are no recommendations regarding future transfusions, although identifying and correcting a positive fluid balance (with diuretics) prior to transfusions is advised.

Nosocomial Infections

The immunosuppressive effects of blood transfusions became evident with the discovery (in the early 1970s) that pre-transplant blood transfusions improve the survival rate of renal allografts (47). Since then, a multitude of clinical studies have shown that *patients who receive blood transfusions have a higher incidence of nosocomial infections* (42,48,49). The risk of infection increases with the volume of blood transfused, and with the storage time of donor blood (49). There is a possibility that patients who receive blood transfusions are sicker and thus more prone to infections, but at least 22 studies have shown that blood transfusion is an independent risk factor for nosocomial infections (49).

What's Worse – Anemia or RBC Transfusions?

A review of 45 clinical studies involving RBC transfusions for anemia in critically ill patients, which included 272,596 patients, revealed the following (50):

- . In 42 of the 45 studies, the risks of RBC transfusions outweighed the benefits, and only one study showed that the benefit of RBC transfusions exceeded the risks.
- . Eighteen studies evaluated the impact of RBC transfusions on survival, and 17 of the 18 studies showed that RBC transfusions were an independent risk factor for a fatal outcome.

Not a very good report card for the use of RBC transfusions to increase the hemoglobin concentration. Wouldn't diuresis do the same thing?

A FINAL WORD

Blood Volume vs. Blood Cells

The practice of boosting hemoglobin levels in critically ill patients is rooted in the belief that anemia is a significant threat to tissue oxygenation. However, it seems that even extreme

reductions in hemoglobin or hematocrit do not impair tissue oxygenation when the intravascular volume is maintained (which allows the cardiac output to increase in response to anemia). Blood volume may have supremacy over the hemoglobin and hematocrit for supporting tissue oxygenation, because hypovolemia is a recognized cause of shock (a state of impaired tissue oxygenation), but anemia is not.

The importance of blood volume is often overlooked in discussions of anemia and RBC transfusions, even by the American Red Cross, whose popular slogan, *blood saves lives*, deserves a more accurate upgrade, as shown in [Figure 12.6](#).

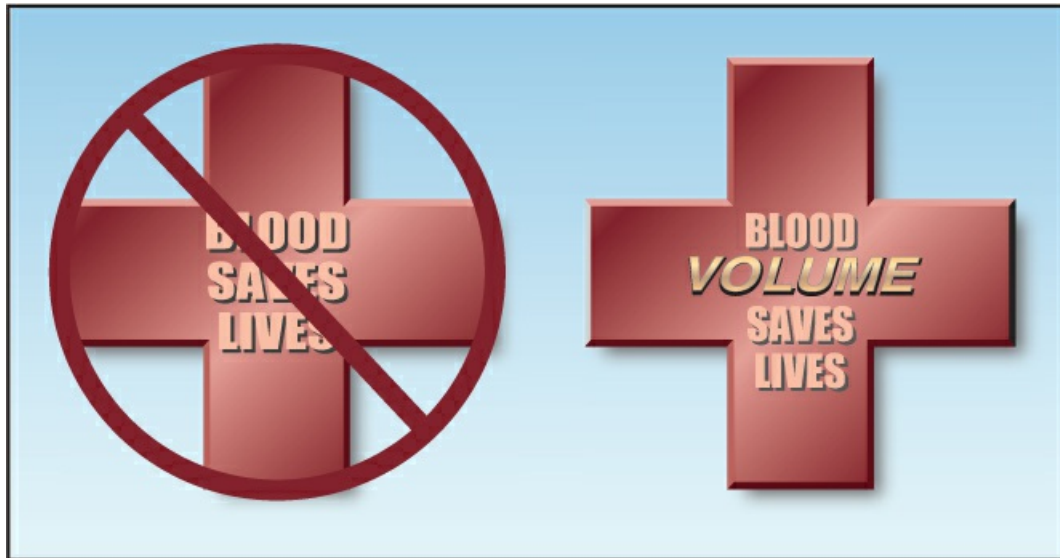


FIGURE 12.6 A popular slogan of the American Red Cross on the left, and an amended version on the right that recognizes the importance of intravascular volume for survival.

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Chapter 13

Platelets and Plasma

Hemorrhage upon a strong pulsation in wounds is bad.

Hippocrates *Aphorisms*, 400 B.C.

The ability to form blood clots is important for limiting blood loss from disrupted blood vessels, which is all the more important if you consider that the human body has a relatively small blood volume (which accounts for only 11–12% of the total body fluids), and loss of as little as 35% of this volume can be fatal (see [Chapter 15](#)). Two of the vital components in the complex pathway to clot formation are the circulating platelets and the procoagulant proteins in plasma. This chapter will focus on the dysfunction of these components in critically ill patients, including the causes, consequences, and corrective actions.

OVERVIEW OF HEMOSTASIS

The vascular endothelium is a thromboresistant surface that is covered with a protective meshwork of mucopolysaccharides called the *glycocalyx* that has two principal functions (1,2). First, it limits the passage of large protein molecules from the plasma, thereby maintaining the colloid osmotic pressure of plasma. Second, it maintains the “fluidity” of blood by preventing leukocytes and platelets from adhering to the endothelial surface, and by acting as a mechanoreceptor for sheer stress that signals endothelial cells to secrete nitric oxide, which promotes vasodilation.

Response to Injury

When the endothelium and its protective glycocalyx is disrupted, platelets adhere to exposed collagen in the subendothelium and begin to form a *platelet plug*, which is the primary stage of hemostasis (3). The platelets release calcium, which activates the glycoprotein receptor IIb/IIIa complex on the platelet surface. This receptor complex binds irreversibly to von Willebrand factor on the surrounding endothelial cells, which helps to anchor the platelet plug to the vessel wall. The IIb/IIIa receptor complex also binds fibrinogen, and the subsequent formation of fibrin bridges between adjacent platelets stabilizes the platelet plug, and eventually leads to its contraction, which promotes the resurgence of blood flow.

Fibrin is essential for the stability of a developing thrombus, and there are two pathways to fibrin formation (4). The major pathway is called the *Tissue Factor Pathway* (formerly called the Extrinsic Pathway), and is activated by the release of thromboplastin from the subendothelium. The second pathway is the *Contact Activation Pathway* (formerly called the Intrinsic Pathway), and is activated by endogenous peptides known as kininogens, which are precursors of bradykinin. Both pathways involve the activation of specific procoagulant proteins known as *clotting factors*, and they both lead to the activation of prothrombin (factor II) and the subsequent conversion of fibrinogen (factor I) to fibrin monomers. This is the secondary stage of hemostasis.

The end-product of the response to injury is the *thrombus*, which is essentially a clump of platelets embedded in a meshwork of fibrin strands and anchored to the vessel wall in the area of injury.

THROMBOCYTOPENIA

Thrombocytopenia is the most common hemostatic disorder in critically ill patients, with a reported incidence as high as 60% (5,6). The traditional definition of thrombocytopenia is a platelet count below 150,000/ μ L, but the ability to form a hemostatic plug is retained until the platelet count falls below 100,000/ μ L (6), so a platelet count <100,000/ μ L is more appropriate for identifying clinically significant thrombocytopenia. However, *the risk of major bleeding is not determined by the platelet count alone*, but also requires a structural lesion that is prone to bleeding. In the absence of such a lesion, platelet counts as low as 5,000/ μ L can be tolerated without evidence of major hemorrhage (7). The major risk with platelet counts <10,000/ μ L is spontaneous intracerebral hemorrhage, which is uncommon (6).

Pseudothrombocytopenia

Pseudothrombocytopenia is a condition where antibodies to EDTA (the anticoagulant in blood collection tubes) produce clumping of platelets in vitro. The clumped platelets are misread as leukocytes by automated machines that perform cell counts, and this results in a spuriously low platelet count. This phenomenon has been reported in 2% of platelet counts performed in hospitalized patients (8).

Suspicion of pseudothrombocytopenia is usually prompted by a platelet count that is lower than expected, or by the presence of clumped platelets on the peripheral blood smear. If suspected, blood collection tubes that use citrate or heparin as an anticoagulant should be used for subsequent platelet counts.

Critically Ill Patients

The most likely causes of thrombocytopenia in the ICU setting are listed in [Table 13.1](#). The sepsis syndrome is the most common cause of thrombocytopenia in ICU patients, which is the result of increased platelet destruction by macrophages (9). Other less common but more life-threatening causes of thrombocytopenia include heparin and the thrombotic microangiopathies; i.e., disseminated intravascular coagulation (DIC), thrombotic thrombocytopenia purpura (TTP), and the hemolytic-uremic syndrome (HUS).

Drugs like antineoplastic agents can produce thrombocytopenia by suppressing platelet

production, but the most common mechanism for drug-induced thrombocytopenia is the production of antibodies that cross-react with platelets (10). This immune-mediated thrombocytopenia is most frequently observed with heparin, and less frequently with platelet glycoprotein receptor (IIb/IIIa) antagonists, and selected antibiotics (particularly linezolid, β -lactams, and vancomycin).

TABLE 13.1 Causes of Thrombocytopenia in the Critically Ill

Nonpharmacological	Pharmacological
Cardiopulmonary Bypass	Anticonvulsants:
Disseminated intravascular coagulation (DIC)	Phenytoin
Hemolytic-Uremic Syndrome	Valproic acid
HIV Infection	Antimicrobial Agents:
Intra-aortic Balloon Pump	β -Lactams
Liver Disease/Hypersplenism	Linezolid
Malignancy	TMP/SMX
Massive Transfusion	Vancomycin
Renal Replacement Therapy	Antineoplastic Agents
Sepsis Syndrome [†]	Antithrombotic Agents:
Thrombotic thrombocytopenia purpura (TTP)	Heparin IIb/IIIa Inhibitors
	Histamine H ₂ Blockers
	Miscellaneous Drugs:
	Amiodarone
	Furosemide
	Thiazides
	Morphine

[†]The most common cause of thrombocytopenia in the ICU setting. From References 5,6,9,10.

HEPARIN-INDUCED THROMBOCYTOPENIA

Heparin-induced thrombocytopenia (HIT) is an immune-mediated condition that is associated with life-threatening thrombosis (arterial or venous) in 25% to 70% of cases (11), and has a mortality rate as high as 30% if left unnoticed (12). The incidence of HIT varies from <1% to 5% (11), and is determined by the type of heparin used and the patient population (see later).

Pathogenesis

Heparin is not immunogenic itself, but it binds to a protein (platelet factor 4) on platelets to form an antigenic complex that can trigger the formation of IgG antibodies. These antibodies bind to platelets and induce a strong platelet activation response that promotes thrombosis. The reticuloendothelial system can clear antibody-coated platelets, which helps to limit the incidence of thrombosis. Heparin-associated antibodies usually disappear within 3 months after discontinuing heparin (12).

Risk Factors

The immune response that triggers HIT *is not a dose-dependent reaction*, and can occur with relatively minor heparin exposures from heparin-based catheter flushes or even heparin-coated catheters (11,13). The risk factors associated with HIT include orthopedic or cardiac surgery, obesity, and exposure to unfractionated heparin; i.e., the risk of HIT with unfractionated heparin (UFH) is 2–10 times greater than the risk with low-molecular-weight heparin (LMWH) (11).

HIT is not common in the general ICU population: i.e., the reported incidence of HIT is <1% in critically ill patients, regardless of the type of heparin used (for DVT prophylaxis) (14). In contrast, HIT has been reported in 1–3% of patients in a cardiothoracic ICU when UFH is used (11).

Clinical Features

HIT typically appears 5 to 10 days after the first exposure to heparin, but it can appear within 24 hours in patients who have had a prior exposure to heparin within the past 3 months (11). Although uncommon, a delayed-onset HIT has been reported that becomes evident up to 3 weeks after exposure to heparin (11). Platelet counts in HIT can fall by 50%, but they rarely fall below 20,000/ μ L.

The feared consequence in HIT is thrombosis, not bleeding. Reports indicate that 17% to 55% of patients with untreated HIT develop venous thromboembolism, and 1% to 3% of patients develop arterial thromboses, which can result in ischemic stroke or acute myocardial infarction (15). *In up to 25% of cases, the thrombosis precedes the thrombocytopenia* (15).

Diagnostic Testing

There are two assays used for the diagnosis of HIT. The first is an enzyme-linked immunosorbent assay (ELISA) for antibodies to the platelet factor 4-heparin complex. This assay has a high sensitivity but a limited specificity, so a negative assay can be used to exclude the diagnosis of HIT, but a positive assay does not confirm the diagnosis (11). The false positive ELISA assays are caused by antibodies to the platelet factor-4 heparin complex that do not activate platelets to produce thrombocytopenia or thrombosis.

The most reliable assay for the diagnosis of HIT is the platelet serotonin-release assay (SRA), which measures the ability of the patient's plasma to activate platelets (which results in enhanced serotonin release by the platelets) (16). This is considered the “gold standard” test for the diagnosis of HIT, and is recommended to confirm the diagnosis of HIT when the ELISA test is positive (16). However, the SRA is not routinely available in many hospitals. As a result, the diagnosis of HIT is usually based on a positive HIT antibody (ELISA) assay.

Risk Assessment

The diagnostic approach to HIT begins by assessing the probability of HIT being present, and a popular risk assessment tool is the 4Ts scoring system shown in Table 13.2 (17). The 4Ts score should be used as follows (18).

- . If the probability of HIT is low on the 4Ts score, then nothing further is needed, including laboratory testing for HIT or discontinuing heparin.
- . If there is an intermediate-risk on the 4Ts score, then heparin should be discontinued and a non-heparin anticoagulant should be initiated. The ELISA immunoassay for HIT antibodies

should be ordered; if the test is negative, the non-heparin anticoagulant can be discontinued, and heparin can be restarted.

- . If there is a high-risk of HIT, then heparin should be discontinued and a non-heparin anticoagulant should be started at full therapeutic doses. This can be discontinued if the ELISA test is negative.
- . Patients with a positive ELISA test for HIT antibodies who have no clinical evidence of thrombosis should have an ultrasound evaluation for deep vein thrombosis in the legs. If a central venous catheter is in place, an ultrasound evaluation for thrombosis in the upper extremities is recommended.
- . All patients with HIT should receive a non-heparin anticoagulant at full therapeutic doses, even if there is no evidence of associated thrombosis. This recommendation is based on studies showing a 10-fold higher incidence of thrombosis following the appearance of HIT when anticoagulation is delayed (13).

TABLE 13.2

The 4Ts Score for Estimating the Risk of HIT

Conditions	Points
Thrombocytopenia:	2
• PLT fall >50% AND nadir ≥ 20 k/ μ L AND no surgery past 3 days	1
• PLT fall 30–50% OR nadir = 10–19 k/ μ L	0
• PLT fall <30% OR nadir <10 k/ μ L	
Timing of onset after heparin exposure:	2
• Onset at 5–10 days OR ≤ 1 day if exposure in past 5–30 days	1
• Possible onset at 5–10 days OR >10 days OR, ≤ 1 day if exposure 31–100 days ago	0
• Onset <4 days without exposure in past 100 days	
Thrombosis or other adverse reactions:	2
• New thrombosis OR skin necrosis OR anaphylactoid reaction	1
• Suspected, progressive, or recurrent thrombosis	0
• None of the above	
Other causes for thrombocytopenia:	2
• None apparent	1
• Possible	0
• Definite	
Scoring: ≤ 3 points = Low risk of HIT (<1%) 4–5 points = Intermediate risk ($\approx 10\%$) 6–8 points = High risk ($\approx 50\%$)	

Alternative Anticoagulants

Note: Warfarin is absolutely contraindicated in patients with acute HIT because it has a prothrombotic effect that can lead to limb gangrene and necrotic skin lesions (11). Vitamin K reversal is indicated for any patient with HIT who has received warfarin within a few days of the diagnosis.

The alternative anticoagulants for patients with HIT include argatroban, bivalirudin, fondaparinux, and the direct oral anticoagulants (DOACs). Argatroban is currently the only drug approved for use in HIT in the United States, but the ASH guidelines for HIT (18) recommends bivalirudin for percutaneous coronary intervention in patients with HIT. These guidelines also state that fondaparinux and the DOACs (primarily rivaroxaban) are reasonable options in patients who are clinically stable.

Argatroban

Argatroban is a direct thrombin inhibitor that is given by continuous infusion. The recommended dosing regimen is as follows (20):

- . Start the infusion at 1–2 µg/kg/min, and titrate dose to achieve an activated PTT of 1.5 to 3 times normal.
- . For hepatic insufficiency, start at 0.5 µg/kg/min.
- . No dose adjustment is needed for renal dysfunction.
- . Full anticoagulation with argatroban should continue until the platelet count rises above 150,000/µL (18).

Long-Term Management

A DOAC can be used for continued anticoagulation after argatroban treatment is completed. Most of the clinical experience with DOACs in HIT have involved rivaroxaban, which has been successful in preventing thrombosis or the extension of thrombosis (18). The duration of treatment is usually 1–3 months. Following an episode of HIT, patients should be advised to wear an emergency identifier for 3 months, and to avoid heparin for life (18).

THROMBOTIC MICROANGIOPATHIES

The thrombotic microangiopathies are life-threatening clinical disorders that share the following features:

- . A consumptive thrombocytopenia.
- . Microvascular thrombosis with dysfunction of one or more major organs.
- . A *microangiopathic hemolytic anemia* that is not immunogenic in origin, but is the result of erythrocyte disruption in the clot-filled microvasculature. The hallmark of this type of hemolytic anemia is the presence of fragmented red blood cells called *schistocytes* on the peripheral blood smear (see Figure 13.1).

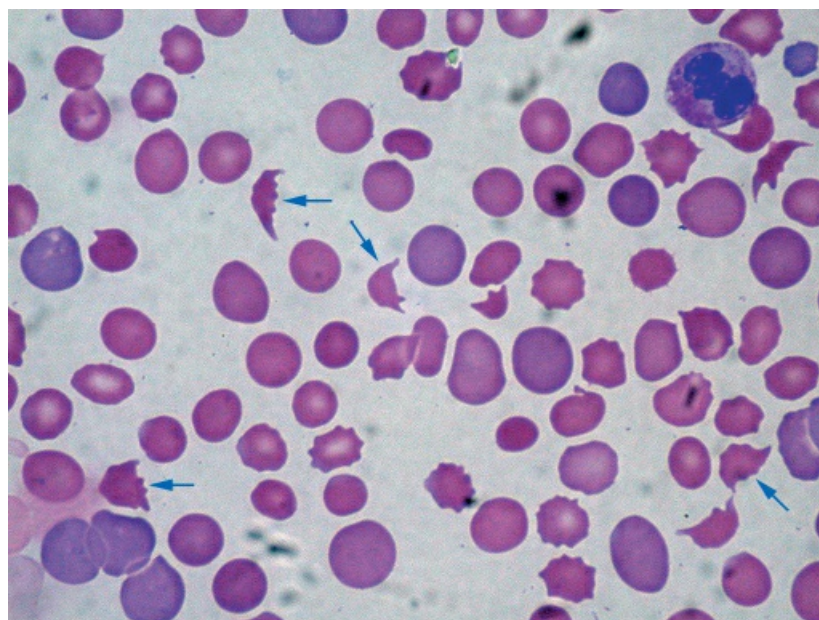


FIGURE 13.1 Peripheral blood smear from a patient with DIC. The arrows are pointing to schistocytes (fragmented erythrocytes) whose presence is pathognomonic of a microangiopathic hemolytic anemia.

There are 3 pathological entities that share these features: disseminated intravascular coagulation (DIC), thrombotic thrombocytopenia purpura (TTP), and the hemolytic-uremic syndrome (HUS). The following is a brief summary of these conditions.

TABLE 13.3 Hematologic Profiles in the Thrombotic Microangiopathies			
Feature	DIC	TTP	HUS
Schistocytes	Present	Present	Present
Platelets	Low	Low	Low
INR	Elevated	Normal	Normal
aPTT	Prolonged	Normal	Normal
Fibrinogen	Low	Normal	Normal
Plasma D-dimer	Elevated	Normal	Normal

From Reference 4. DIC = disseminated intravascular coagulation, TTP = thrombotic thrombocytopenia purpura, HUS = hemolytic uremic syndrome.

Disseminated Intravascular Coagulation

Disseminated intravascular coagulation (DIC) is a secondary disorder that is triggered by conditions that produce widespread endothelial and tissue injury such as multisystem trauma, septic shock, and obstetric emergencies (amniotic fluid embolism, abruptio placentae, eclampsia, and retained fetus syndrome). The inciting event is release of *tissue factor*, which activates a series of clotting factors in the bloodstream that culminates in the formation of fibrin. This leads

to widespread microvascular thrombosis and secondary depletion of platelets and clotting factors, resulting in a *consumptive coagulopathy* (21).

Clinical Features

The microvascular thrombosis in DIC can lead to multiorgan dysfunction, most often involving the lungs, kidneys, and central nervous system, while depletion of platelets and coagulation factors can promote bleeding. Thrombosis is most prominent when sepsis is the underlying cause, while bleeding is prominent in the obstetric conditions. A severe form of DIC known as *purpura fulminans* is characterized by necrosis and gangrene of the limbs, and is seen with overwhelming infections, most notably with meningococemia (22).

HEMATOLOGIC ABNORMALITIES: The hematologic features of DIC are summarized in Table 13.3. The hemolytic anemia is suspected by an elevated LDH and reduced haptoglobin level, and is confirmed by the presence of schistocytes, like the ones in Figure 13.1. What distinguishes DIC from the other thrombotic microangiopathies is the depletion of clotting factors, which prolongs the INR and the activated partial thromboplastin time (aPTT). Fibrinolysis is also enhanced, which elevates the fibrin degradation products in plasma (i.e., plasma D-dimers).

Management

There is no effective treatment for DIC, and management involves supportive care and treatment of the underlying cause. Life-threatening hemorrhage prompts replacement therapy with blood products, but this usually does not stop the bleeding, and the addition of platelets and coagulation factors can “add fuel” to the microvascular thrombosis. When DIC is accompanied by multiorgan failure, the mortality rate is $\geq 80\%$ (21).

Thrombotic Thrombocytopenia Purpura

Thrombotic thrombocytopenia purpura (TTP) is a potentially devastating condition that is treatable, but can be fatal in 24–48 hours if left untreated.

Pathogenesis

One of the components of the hemostatic process is the von Willebrand factor (VWF), a large protein from endothelial cells that promotes platelet adhesion. The activity of VWF is normally controlled by a protease enzyme with the name of ADAMTS-13 that cleaves VWF and reduces its prothrombotic activity. The principal problem in TTP is the appearance of antibodies that inactivate the ADAMTS-13 protease (22), resulting in uncontrolled platelet adhesion and widespread microvascular thrombosis. The trigger for these antibodies is not known.

Clinical Features

TTP usually appears in middle age, and is three times more frequent in women. The classic presentation of TTP is a *pentad of manifestations* that include fever, altered mentation, renal insufficiency, thrombocytopenia, and microangiopathic hemolytic anemia. All 5 of these may not be present at the outset; i.e., fever may be absent, and neurologic symptoms may be mild (e.g., headache) or absent. However, this condition can progress rapidly to multiorgan failure and a fatal outcome.

Diagnosis

Prompt diagnosis of TTP is important because delays in treatment can diminish the chances of survival. A marked decrease in ADAMTS-13 activity in plasma is patho-gnomonic of TTP (22), but this test is not readily available in most hospitals. The presence of thrombocytopenia and schistocytes will signal the presence of a thrombotic microangiopathy. TTP can then be distinguished from DIC because *the INR, aPTT, and fibrinogen levels are normal in TTP* (see Table 13.3), and it can be distinguished from the hemolytic uremic syndrome because the renal insufficiency in TTP is often mild (i.e., the creatinine is usually <2 mg/dL) (22).

Treatment

The treatment of choice for TTP is plasma exchange (23), where blood from the patient is diverted to a device that separates and discards the patient's plasma and reinfuses the patient's erythrocytes with plasma from a healthy donor. About 1.5 times the normal plasma volume is exchanged with each treatment, and this is continued daily until the platelet count returns to normal for 2 consecutive days. Acute fulminant TTP is almost always fatal if untreated, but if plasma exchange is started early (within 48 hours of symptom onset), as many as 90% of patients can survive the illness (23,24).

Hemolytic Uremic Syndrome

As the name indicates, the hemolytic uremic syndrome (HUS) is characterized by the combination of a microangiopathic hemolytic anemia and acute renal failure, in addition to a consumptive thrombocytopenia. HUS can be the consequence of a variety of disorders, but infection is probably the most common predisposing condition, especially infections caused by *Escherichia coli* strains that produce the Shiga toxin, and by *Streptococcus pneumoniae* and the influenza virus (25). Other predisposing conditions include pregnancy, autoimmune disorders (e.g., lupus), the antiphospholipid syndrome, malignancy, and drugs (e.g., quinine, cyclosporin, tacrolimus). There is also an inherited condition known as *atypical HUS* that is characterized by unregulated complement activation (26).

Clinical Features

HUS presents with the triad of thrombocytopenia, acute kidney injury and a microangiopathic hemolytic anemia. Extrarenal manifestations are reported in about 20% of cases of atypical HUS (26), and most frequently are neurologic, ranging from agitation to coma.

The diagnosis of HUS requires evidence of a microangiopathic hemolytic anemia (first suspected by elevated LDH and reduced haptoglobin levels, and then confirmed by the presence of schistocytes). HUS differs from DIC in that there is no consumption of clotting factors in HUS, so the INR and aPTT are normal in HUS. Fibrinolysis is also not enhanced, so the products of fibrinolysis (i.e., D-dimer levels) are not increased (see Table 13.3). HUS can be differentiated from TTP by the severity of the renal insufficiency, which is usually greater in HUS (22).

Management

The management of infection-related HUS involves supportive care and treatment of the underlying infection. In cases of atypical HUS due to unregulated complement activation (see earlier), a monoclonal IgG antibody named eculizumab (which binds to the C5 complement

protein and blocks the prothrombotic effects of complement activation) has had success in reversing the condition (27). If this drug is not available, then plasma exchange is the treatment of choice.

Management of the renal failure in HUS involves the usual measures, including renal replacement therapy if needed. Renal transplantation has been used in atypical HUS.

PLATELET TRANSFUSIONS

Indications

Note: The following recommendations for platelet transfusions do not apply to cases of consumptive thrombocytopenia in the thrombotic microangiopathies (where platelet transfusions can aggravate the microvascular thrombosis).

Active Bleeding

In the presence of active bleeding, platelet transfusions are recommended to keep the platelet count above 50,000/ μ L (28). For intracranial hemorrhage, the platelet count should be kept above 100,000/ μ L (28).

Prophylactic Transfusions

The most recent guidelines from the American Association of Blood Banks (AABB) includes the following recommendations for prophylactic platelet transfusions (29):

- . In the absence of bleeding, platelet transfusions are recommended when the platelet count is $\leq 10,000/\mu\text{L}$.
- . For procedures, the following platelet counts are recommended as thresholds for prophylactic platelet transfusions: 1) $<50,000/\mu\text{L}$ for elective non-neuraxial surgery, 2) $<50,000/\mu\text{L}$ for lumbar punctures, and 3) $<20,000/\mu\text{L}$ for insertion of central venous catheters.

One aspect of these recommendations that deserves mention is that none of them have been validated (e.g., platelet counts as low as 5,000/ μ L do not provoke spontaneous bleeding, as mentioned earlier). Overall, it is reasonable to state that, *in the absence of active bleeding, the major problem with thrombocytopenia is not the platelet count, but the underlying cause of the thrombocytopenia.*

Platelet Preparations

Platelets are provided either as pooled platelets extracted from multiple units of donated whole blood, or by extracting platelets from a single donor using apheresis techniques. Platelets are stored at room temperature, which limits their shelf life to about 5 days.

Pooled Platelet Concentrates

Platelets are separated from fresh whole blood by differential centrifugation, and the resulting platelet concentrates from multiple (4–6) donors are pooled together prior to storage. The pooled platelet concentrate contains about 38×10^{10} platelets in 260 mL plasma, which is equivalent to a

platelet count of about $130 \times 10^9/\mu\text{L}$. This is six orders of magnitude higher than the normal platelet count in blood ($150\text{--}400 \times 10^3/\mu\text{L}$).

Apheresis Platelets

Apheresis platelets are collected from a single donor, and have a platelet count equivalent to the pooled platelet concentrates. The presumed benefit of single-donor platelet transfusions is a lower risk of transmitted infections and a lower incidence of platelet *alloimmunization* (i.e., developing antibodies to donor platelets, which reduces the effect of platelet transfusions). However, neither of these proposed benefits has been validated (30,31) and in a recent national survey that included over 2 million platelet transfusions (31), adverse reactions were much more common with single-donor (apheresed) platelets compared to multiple-donor preparations (see Table 13.4). In addition, there is no difference in the risk of platelet alloimmunization with single-donor and multiple-donor transfusions when the platelet preparations are “leuko-reduced” (32).

Leukoreduction

Leukocytes in donor blood have been implicated in several adverse reactions to transfusion, and leukocyte removal using specialized filters is now a routine practice for erythrocyte transfusions (see last chapter). Platelet concentrates are not free of leukocytes, and leukocyte reduction in platelet preparations reduces the incidence of febrile reactions, as well as cytomegalovirus transmission (which is transmitted in leukocytes) and platelet alloimmunization (28,32). Because of these advantages, leukocyte reduction is becoming a routine practice for platelet transfusions.

Response to Transfused Platelets

In an average-sized adult with no ongoing blood loss, *the infusion of one unit of platelets (either multiple-donor or apheresed platelets) should raise the circulating platelet count by 35,000 to 50,000/ μL at one hour post-transfusion (26)*. The increment is about 40% lower after 24 hours, as shown in Figure 13.2 (33).

The increment in platelet count declines with multiple transfusions. This is also shown in Figure 13.2, where the platelet increment is about 25% lower after 5 platelet transfusions (33). This phenomenon of “platelet refractoriness” is the result of antiplatelet antibodies in the recipient directed at ABO antigens on donor platelets. This effect is mitigated by transfusing ABO-matched platelets.

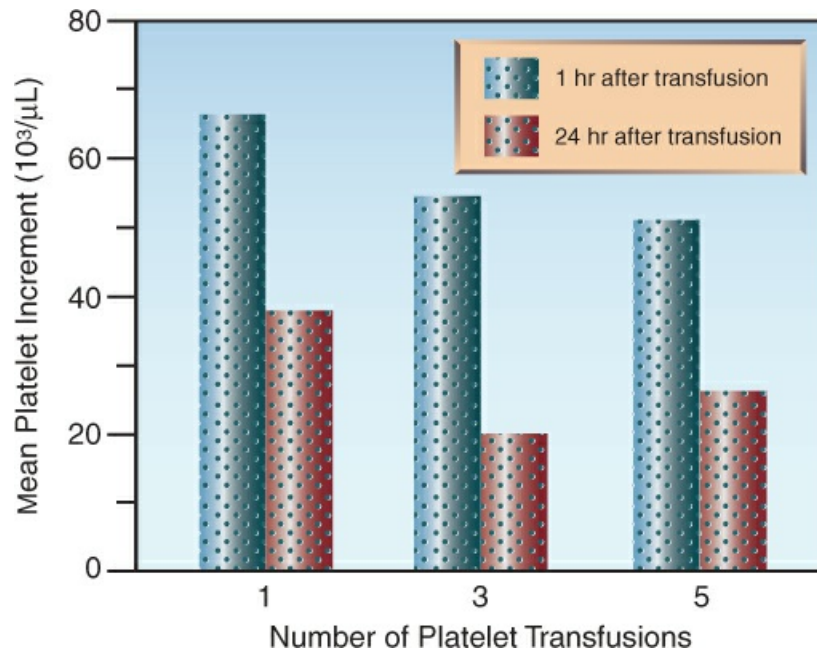


FIGURE 13.2 Post-transfusion increment in platelet counts in relation to the time elapsed after the transfusion (one hour versus 24 hours) and the number of transfusions given. Data from Reference 33.

Adverse Effects

The rate of adverse reactions to platelet transfusions is shown in [Table 13.4](#). The data in this table is from a survey of more than 2 million platelet transfusions, and compares the reaction rates with single-donor and multiple-donor preparations (31). Note the low overall reaction rate, and the higher rate with single-donor preparations. The following are some relevant points about these reactions.

Nonhemolytic Fever

Transfusion-associated fever can be the result of antileukocyte antibodies in recipient blood that react with antigens on leukocytes in donor blood. This triggers the release of endogenous pyrogens from phagocytes, which is the source of the fever. The routine use of leukocyte reduction in platelet preparations has significantly reduced (but not eliminated) this reaction.

Allergic Reactions

Allergic reactions (urticaria, anaphylaxis) are the result of sensitization to plasma proteins in donor blood. Single-donor platelets have a higher plasma volume than multiple-donor preparations, which would explain the higher rate of allergic reactions with single-donor platelets in [Table 13.4](#).

Acute Lung Injury

Transfusion-related acute lung injury (TRALI) is described in [Chapter 12](#). This condition is the result of antileukocyte antibodies in donor plasma that bind to antigens on circulating neutrophils in the recipient. The result is a diffuse, inflammatory lung injury that is similar to the acute respiratory distress syndrome (ARDS). This reaction is most often associated with erythrocyte

transfusions, but also occurs with platelet transfusions.

Transmission of Infections

The adoption of nucleic acid testing has virtually eliminated the risk of transmission of HIV and the hepatitis viruses in both erythrocyte and platelet transfusions. The risk of bacterial transmission is higher with platelet than erythrocyte transfusions because platelets are stored at room temperature, which is more likely to promote bacterial proliferation. However, the rate of bacterial transmission with platelet transfusions is quite low, as shown in [Table 13.4](#).

TABLE 13.4 Adverse reactions to Different Platelet Preparations (Rate per 100,000 Units Transfused)			
Adverse Reaction	Multiple-Donor Preparations	Single-Donor Preparations	p value
Nonhemolytic Fever	37	136	<0.01
Allergic Reaction	27	325	<0.01
Acute Lung Injury	1	22	<0.05
Circulatory Overload	7	8	NS
Acute Hemolysis	0.2	1.8	<0.01
Bacterial Transmission	0.8	1.4	NS
Viral Transmission	0	0	NS
Total Reactions	70	478	<0.01

From Reference 31. NS = not significant.

PLASMA PRODUCTS

Plasma products are used as a source of procoagulant plasma proteins, and the two plasma products used most often for this purpose are fresh frozen plasma and cryoprecipitate.

Fresh Frozen Plasma

Plasma is separated from donor blood and frozen at -18°C within 8 hours of blood collection. This *fresh frozen plasma* (FFP) has a volume of about 230 mL, and can be stored for one year. Once thawed, FFP can be stored at $1\text{--}6^{\circ}\text{C}$ ($34\text{--}43^{\circ}\text{F}$) for up to 5 days. The principal uses of FFP include the resuscitation of massive blood loss, and the reversal of anticoagulation with warfarin.

Massive Blood Loss

The current practice for the resuscitation of massive blood loss (i.e., loss equivalent to the blood volume within 24 hours) is to transfuse one unit of FFP for every 1–2 units of packed red blood cells (34). There are two drawbacks with FFP: i.e., the time to thaw FFP is a drawback when immediate availability is needed for the resuscitation effort, and the limited shelf-life after thawing can lead to wasting the product. These limitations have led to the introduction of “liquid

plasma” preparations that are never frozen, and can be stored 1–6° C (34–43° F) for up to 26 days. Early studies of liquid plasma have shown equivalent efficacy with FFP, while also reducing the waste associated with FFP (35).

Prophylactic FFP

A majority of FFP infusions in ICU patients are used to reduce an elevated INR in coagulopathic, nonbleeding patients who are considered at risk for bleeding (e.g., cirrhotic patients with a prior history of variceal bleeding) (36). However, *there is no evidence that prophylactic FFP reduces bleeding* in these patients, or that it provides any type of clinical benefit (37,38).

Reversing Warfarin Anticoagulation

One of the traditional uses of FFP is to replenish clotting factors in cases of troublesome bleeding related to anticoagulation therapy with warfarin. Vitamin K is given first to block ongoing anticoagulant activity, and the FFP is then given (usually in a volume of 10–15 mL/kg) to return the INR to normal (39). This approach has two shortcomings: i.e., the time needed to thaw the FFP leads to treatment delays (which can prolong the bleeding), and the volume of FFP that is required can aggravate the bleeding. These problems have been eliminated by the introduction of *4-factor prothrombin complex concentrate*, a lyophilized powder that contains the vitamin K-dependent clotting factors (factors II, VII, IX, and X) and, with a reconstituted volume of 100 mL, is capable of reversing warfarin effects in less than 30 minutes (40). (See [Chapter 47](#) for more on warfarin reversal.)

FFP and the INR

There is a fundamental problem with the use of the INR to guide FFP infusions. (*Note:* INR is the abbreviation for “international normalized ratio”, which is the ratio of the patient’s prothrombin time divided by a standardized normal prothrombin time.) This test was introduced to monitor anticoagulation with coumadin, and it is heavily influenced by the activity of the vitamin K-dependent clotting factors (Factors II, VII, IX, and X). However, the INR does not monitor other components of hemostasis, such as the activity of endogenous “anticoagulant” proteins (i.e., antithrombin, protein C, and protein S), which must be considered to determine the overall state of hemostasis.

The graphs in [Figure 13.3](#) show how the INR can be misleading as a guide for FFP infusions (38). The data in these graphs is from a study of prophylactic FFP infusions (12 mL/kg) for an elevated INR in nonbleeding ICU patients who were scheduled for a procedure. Note that the FFP infusions significantly reduced the INR (graph on the left), but they also significantly increased the activity of antithrombin (graph on the right), an anticoagulant protein. (Although not shown, the activity of the other anticoagulant proteins was increased after the FFP infusion.) Thus, despite the favorable effect on the INR, there may be little or no improvement in overall coagulation status because of the opposing effect on endogenous anticoagulant activity.

Adverse Reactions

The adverse reactions to plasma transfusion are similar (in type, not frequency) to the reactions that complicate erythrocyte and platelet transfusions.

Acute Hemolytic Reactions

Acute hemolytic reactions are caused by anti-A and anti-B antibodies in donor plasma that react with A and B antigens on recipient RBCs. Cross-matching of plasma transfusions for ABO compatibility has become a standard practice, which eliminates the risk of acute hemolytic transfusion reactions.

Nonhemolytic Fever

Fever that is unrelated to acute hemolysis is caused by leukocytes in donor blood, which theoretically should not be a risk with plasma transfusions. However, FFP is not free of leukocytes, and this creates a minimal risk of nonhemolytic fever; i.e., in one report, the risk was <1 per 100,000 transfusions (41).

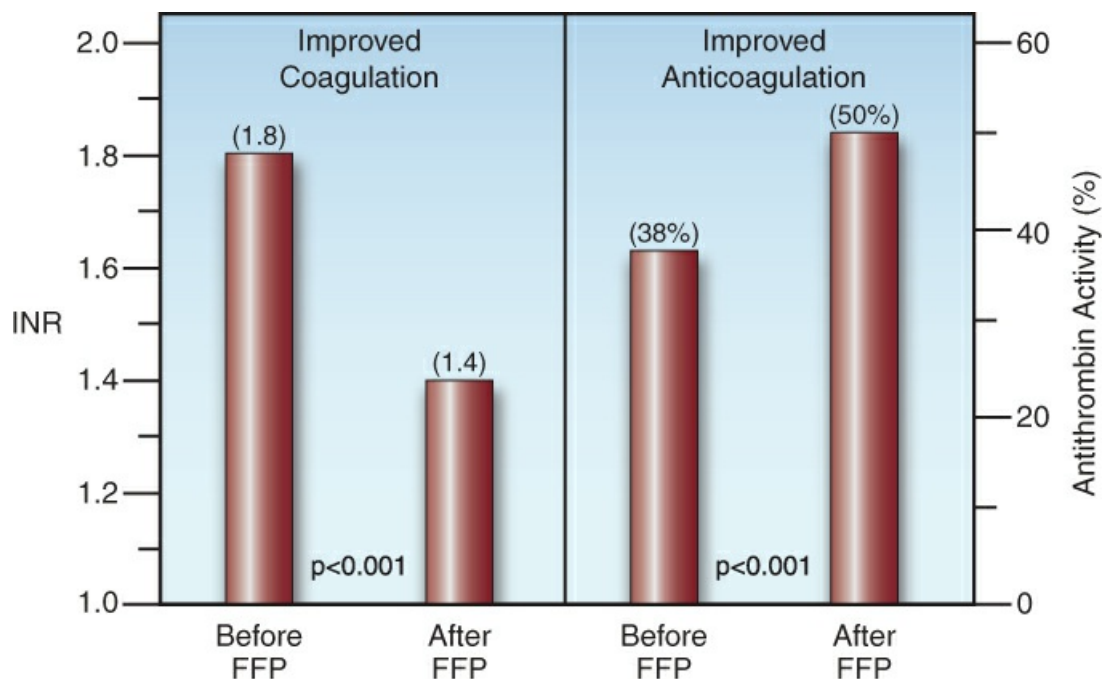


FIGURE 13.3 The influence of FFP (12 mL/kg) on the INR (graph on the left) and the activity of antithrombin (graph on the right) in nonbleeding ICU patients. Height of the bars indicate median values. Despite the favorable decrease in INR, the opposing effect on antithrombin could result in little or no change in coagulation status. Data from Reference 38.

Hypersensitivity Reactions

Hypersensitivity reactions (urticaria, anaphylaxis, anaphylactic shock), which are caused by sensitization to proteins in donor plasma, are more common with plasma transfusions than with erythrocyte or platelet transfusions. The reported incidence of allergic reactions to FFP ranges from 1:600 to 1:2,000 transfusions (41).

Acute Lung Injury

Transfusion-related acute lung injury (TRALI) is attributed to antileukocyte antibodies in donor blood, and is a complication of erythrocyte, platelet, and plasma transfusions. The reported incidence of TRALI with plasma transfusions is about 1:250,000 units transfused (41). The clinical features of TRALI are described in Chapter 12.

Hydrostatic Pulmonary Edema

Transfusion-associated circulatory overload (TACO) is considered a risk if the infusion rate of FFP exceeds 1 mL/kg per hour in patients who are predisposed to circulatory overload (e.g., those with heart failure or pre-existing fluid overload). The reported risk of TACO ranges from 1:70 to 1:1500 units transfused (42).

Transmitted Infections

Plasma transfusions carry a minimal risk of transmitting infections. The risk of hepatitis B transmission is 1:280,000 transfusions, the risk of hepatitis C transmission is 1:1.2 million transfusions, and the risk of HIV transmission is 1:1.6 million transfusions (41). The risk of bacterial transmission is reported as “rare” and CMV transmission, which occurs in transfused leukocytes, has not been reported with plasma transfusions (41).

Cryoprecipitate

When FFP is allowed to thaw at 4° C, a milky precipitate forms that is rich in cold-insoluble proteins (cryoglobulins) like fibrinogen, von Willebrand factor, and factor VIII. This *cryoprecipitate* can be separated from plasma and stored at –18° C for up to one year. It has a fibrinogen concentration of 10–20 g/L, and a storage volume of 10–15 mL. Once thawed, cryoprecipitate must be used within 4 hours (43).

Using Cryoprecipitate

Cryoprecipitate can be used as a source of fibrinogen in massive blood loss from trauma or postpartum hemorrhage. Fibrinogen depletion is always a consideration in the setting of significant blood loss because it is the earliest clotting factor to become depleted, and is also susceptible to the dilutional effects of fluid resuscitation. For major blood loss in trauma, cryoprecipitate is recommended when the fibrinogen level falls to 150–200 mg/dL (44). One unit of cryoprecipitate per 10 kg body weight will increase the plasma fibrinogen level by about 50 mg/dL (43), but this is in the absence of bleeding, so expect a greater cryoprecipitate requirement during active hemorrhage. Alternately, cryoprecipitate administration can be guided by viscoelastic assays, such as thromboelastography (which is described in Chapter 15).

Cryoprecipitate has some drawbacks, including a variable fibrinogen concentration, the time needed for thawing (which can delay infusion and prolong bleeding), and an increased risk of transmitted infections (because it is not subjected to pathogen-reduction measures) (45). These limitations are not shared by *fibrinogen concentrates*, which have a consistent fibrinogen content of 20 g/L, and do not require thawing; as a result, fibrinogen concentrates are gaining popularity as a superior alternative to cryoprecipitate (45).

A FINAL WORD

Changing Perceptions

There are two general messages in this chapter that might change your perception of thrombocytopenia in critically ill patients:

- . In the absence of active bleeding, the problem with thrombocytopenia is not the platelet count, but the underlying illness. As a result, prophylactic platelet transfusions are rarely necessary in critically ill patients.
- . The most life-threatening consequence of thrombocytopenia is not bleeding, but thrombosis (as is evident in HIT, DIC, TTP and HUS).

And there is one general message that might change your perception of the INR as a measure of coagulation status:

- . The INR is sensitive to the activity of vitamin K-dependent coagulation factors, but it does not monitor other components of the hemostatic process, such as the activity of endogenous anticoagulant proteins (i.e., antithrombin, Protein C and Protein S). This limitation affects the reliability of the INR as a gauge for FFP infusions; i.e., a decrease in the INR in response to FFP is considered evidence of an improvement in the ability to form a clot, but FFP also increases the activity of endogenous anticoagulant proteins, and the net result may be little or no change in clot-forming ability. Therefore, the INR provides misleading information for FFP infusions.

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SHOCK SYNDROMES

*In the successful resuscitation of the shocked patient,
the physician achieves his or her greatest victory.*

Evan Geller,
1993